

spec

Feature Specification: Authored AI/IT Knowledge Areas with Lessons, Tests and Video Anchors

Feature Branch: main (no feature branch; this repository works on main only)

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Status: Draft

Input: User description: “Workshop curriculum must publish authored AI/IT knowledge areas — not mined transcript vocabulary — each with lessons, senior-level interview question banks and an end-of-area test, rich materials (illustrations, diagrams, schemes, graphs, and video with time ranges), video deep-links that seek the recording and scroll the transcript to that position, tags and every field ai_interviewing exposes, a vivid accessible theme derived from seeded tokens, and full standalone cloneability. Measured defects to resolve: 814 of 819 published areas are single mined terms (3.9% AI-relevant); the list serves 819 areas while the detail route refuses 817 with area_not_published; area detail pages render raw ULIDs and have no lessons or tests; the widened palette never reached the served stylesheets. Every requirement must be covered by machine-produced deterministic evidence with three-valued gates.”

Why this specification exists

This is not a greenfield feature. It is the remedy for a set of defects an operator found by **manual retesting of the running platform**, each of which has since been reproduced by machine and is quoted below

with its measurement. The specification exists so the remedy is designed once, in the open, rather than patched four times by four agents.

Nothing in this document quotes private content. The curriculum corpus, the recordings and the notes derived from them live in a PRIVATE submodule while this specification lives in a PUBLIC repository. Areas, terms, questions and transcript text are referred to by **shape, count and location only** — never by their words. A single pasted line would be a permanent disclosure.

The measured starting state

#	Finding	Measurement
D1	Knowledge areas are mined vocabulary , not authored subjects	814 of 819 published areas carry a single extracted term. Two independent probes agree the set is not topical: a topical classifier scored 3.9% AI/IT-relevant, and a term-in-name probe scored 2.4% carrying an AI/ML term and 4.5% carrying an AI or IT term. No relevance threshold rescues the set — at classifier floors 2 and 3 the surviving population is 0 , and at an evidence floor of 2 the AI-term share falls to 0 of 36 .
D2	Area identity is machine-readable only	Area detail pages rendered the raw opaque identifier as the page heading in 3 of 4 render branches.
D3	The catalogue and the detail route disagree	The list endpoint serves 819 areas; the detail route refuses

#	Finding	Measurement
D8	A verified on-disk state is not a served state	817 of them as not published. A visitor can therefore reach a listed area and be told it does not exist. With the corrected catalogue verified on disk, the running server continued to serve the old 819 because it had loaded its taxonomy at start-up. The served result was undetermined until the process reloaded. This is D6's failure mode on a second surface.
D4	There are no lessons and no tests	The lessons and assessment routes return not-found for every area.
D5	Tags are discarded before the interface can ask for them	The list response carries tags for every row; the client-side area model has no field for them at all.
D6	The palette work never reached the visitor	Token files widened the hue span, but no served stylesheet was regenerated from them. A token-level measurement is evidence about tokens — it is not evidence about the interface.
D7	Each module cannot be used on its own	The two private learning modules could not be cloned and run outside this

#	Finding	Measurement
		umbrella without hand-assembling their dependencies.

Clarifications

Session 2026-09-07

These five decisions were the highest-impact unresolved points in the ambiguity scan. **Each was resolved from measurement taken on this tree, not from preference**, and each is recorded with the evidence that decided it so an operator can overturn it by disputing the evidence rather than the taste. The measurements arrived from the area-authoring work that ran concurrently with this specification; nothing here is speculative, and nothing here is a completion claim.

- Q: How is the authored area set bounded — how many areas, and what decides which subjects the recorded material warrants? → A: The **37 modules of the existing hand-authored lexicon** (7 tracks, 137 terms), each published only with measured passage evidence, an authored document and a review. Not a new mined list, and not an evidence-frequency ranking.
- Q: Must a learner have a durable identity for test gating? → A: **No — anonymous, per-browser**. Gating is enforced on the serving side by withholding the question bank, not by trusting the client.
- Q: May assessment questions be drafted from the recorded material, or must they be written independently? → A: **Authored against this corpus only**. Framing and structure may be reused from the established interviewing module; questions themselves may not be imported.
- Q: What must be true before an interface measurement counts as evidence about the served product? → A: The served catalogue **MUST declare the corpus revision it was built from**, and an interface claim **MUST** cite it.
- Q: What is the pass threshold for an end-of-area test? → A: **A stated per-area threshold, defaulting to 70%**, shown to the learner before they start.

The evidence behind each

Q1 — the area set is 37, and frequency was measured and rejected as the selector. A hand-authored lexicon already existed in the private corpus and was unused by the publication path: **7 tracks, 37 modules, 137 terms**, with alias patterns written against what the transcription engines actually emitted rather than against idealised spellings. Measured across all three recordings: **0 of 37** modules have zero evidence and **0 of 137** terms match zero passages, so the set is fully evidenced. It is not evenly evidenced — **16** modules are strongly evidenced (≥ 40 matching passages), **9** moderately (20–39), and **12** are thin (< 20), the thinnest at exactly **1** matching passage in the entire corpus.

An evidence floor was tried as the selector and **rejected by measurement, because it made relevance worse**: at floor 2, **0 of 36** surviving titles carried an AI/ML term, against **32 of 814** at floor 1. Frequency is not topicality. The set is therefore bounded by authored subject matter with evidence *reported*, not by evidence *ranked*.

Q2 — anonymous is sufficient because the gate is not on the client. No sign-in exists. A per-browser record is enough to decide what to *show*, because the security property that matters is stated in FR-010 and SC-009: the serving side does not emit question prompts, options or answers for an unavailable test. A tampered client can therefore reveal only that it has tampered — it cannot reveal an answer key it was never sent. Durable cross-device progress would be a separate capability with its own identity requirements.

Q3 — the prohibition is written down, and it was hit during authoring. The private module's own knowledge-model contract forbids importing questions from the interviewing module. The authoring work resolved this the way both documents permit: reuse the framing, author every question and scenario against this corpus, and record the decision per area. Where the recordings did not cover a topic the interviewing module covers, the port was **refused and the refusal named** rather than filled with plausible content.

Q4 — this was measured as a live discrepancy, not anticipated as a risk. With the corrected catalogue verified on disk, the running server still served the **old 819** areas, because it had loaded its taxonomy at start-up. The on-disk state was verified; the served state was **undetermined** until the container reloaded. That is precisely defect D6

in a second costume — a measurement of the source reported as a measurement of the product. A declared corpus revision on the served catalogue makes the discrepancy detectable instead of invisible.

Q5 — 70% is a default, and the number is less important than its visibility. The requirement is that the threshold is stated per area and shown before the attempt. An undisclosed threshold makes a result unfalsifiable to the learner.

User Scenarios & Testing (*mandatory*)

User Story 1 - A learner finds a subject they recognise (Priority: P1)

A learner opens the knowledge catalogue and sees a list of named AI and IT subjects — each with a human-readable title, a one-paragraph summary of what it covers, and tags describing its domain and difficulty. Every entry they can see is an entry they can open.

Why this priority: Every other story in this document is reached *through* this list. While the catalogue shows opaque identifiers and refuses to open most of them, no other capability is reachable, so nothing else can be demonstrated. This story alone — a catalogue that is honest about what it holds — is a viable increment.

Independent Test: Open the catalogue, read every visible entry, and open each one. Delivers value if every entry is nameable by a person and every entry opens.

Acceptance Scenarios:

1. **Given** the catalogue is served, **When** a learner views the area list, **Then** every entry shows a title composed of words rather than an opaque identifier.
 2. **Given** the catalogue is served, **When** a learner opens any entry from the list, **Then** the detail view for that entry loads and never reports that the entry does not exist.
 3. **Given** the catalogue is served, **When** a learner reads any entry, **Then** it shows a summary and at least one tag.
 4. **Given** an area exists but is not ready to be shown, **When** the catalogue is assembled, **Then** it does not appear in the list at all — being listed and being openable are the same condition.
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User Story 2 - A learner works through an area and is tested on it (Priority: P1)

Having opened an area, the learner reads its lessons in order. At the end of the area they take a test built from senior-level interview questions on that subject, submit answers, and receive a result telling them which questions they got right and what the correct answers were.

Why this priority: This is the mission the operator stated — areas covered “with lessons and senior interviewing questions and tests”. An area that opens but teaches nothing is the defect D4 restated, not a fix for it. It shares P1 with Story 1 because a catalogue of empty areas delivers no learning value.

Independent Test: Open one area, read its lessons end to end, take its test, answer deliberately (some right, some wrong), and check that the score and the per-question feedback match what was answered.

Acceptance Scenarios:

1. **Given** a published area, **When** the learner opens it, **Then** they see its lessons listed in a stated order, each with a title and an estimated reading time.
 2. **Given** a published area, **When** the learner opens a lesson, **Then** they see its body content and can move to the next and previous lesson without returning to the area.
 3. **Given** an area whose lessons the learner has not finished, **When** they look for its test, **Then** they are told what remains before the test opens, and the test’s questions and answers are not disclosed.
 4. **Given** an area whose lessons the learner has finished, **When** they take the test, **Then** they may answer each question, submit once, and receive a score plus, for every question, whether they were right and what the correct answer is.
 5. **Given** a submitted test, **When** the learner re-reads the result, **Then** the result is the same as when it was first shown.
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User Story 3 - A learner jumps from a lesson to the exact moment in the recording (Priority: P2)

A lesson references a moment in a recorded session. The learner follows the reference and lands on the recording positioned at that moment, with the accompanying transcript scrolled to the matching passage and that passage marked so it is findable on the page.

Why this priority: This is the capability that makes the lessons worth more than prose — it binds authored teaching to the primary recorded material. It sits below P1 because a learner can complete Stories 1 and 2 without it, and it depends on Story 2 existing to link *from*.

Independent Test: Follow every reference in one area's lessons and confirm each lands on the right recording, at the right moment, with the right passage in view and marked.

Acceptance Scenarios:

1. **Given** a lesson with a reference to a recorded moment, **When** the learner follows it, **Then** the recording page opens with the player positioned at the referenced start time.
 2. **Given** the same reference, **When** the page settles, **Then** the transcript is scrolled so the referenced passage is visible without further scrolling, and that passage is visually distinguished from its neighbours.
 3. **Given** a reference that names a start and an end, **When** the learner follows it, **Then** the extent of the referenced range is discernible, not only its start.
 4. **Given** a reference whose target recording or passage is not available, **When** the learner follows it, **Then** they are told the target is unavailable — they are never sent to a page that reports nothing found.
 5. **Given** any published lesson, **When** its references are inspected, **Then** every one of them resolves to an existing recording and an existing passage.
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User Story 4 - A learner uses supporting materials, not only text (Priority: P2)

Within a lesson the learner encounters illustrations, diagrams, schemes and graphs alongside the prose, and — where the subject warrants it — a segment of recorded video presented in place rather than as a link away.

Why this priority: The operator named these explicitly. They raise comprehension for the same content, but a lesson without them still teaches, so they rank below the lesson mechanism itself.

Independent Test: Open every lesson in one area and confirm each declared material renders, carries a caption, and is reachable without a mouse.

Acceptance Scenarios:

1. **Given** a lesson declaring a visual material, **When** the learner reads it, **Then** the material renders in place with a caption stating what it shows.
 2. **Given** a lesson declaring a video segment, **When** the learner reads it, **Then** the segment is playable in place and bounded to its stated range.
 3. **Given** any material, **When** it is presented to assistive technology, **Then** it carries a text alternative that conveys its content rather than its file name.
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User Story 5 - The interface is vivid, legible and consistent (Priority: P2)

A visitor perceives the platform as a designed product with a distinct and varied palette, rather than a two-tone monochrome document — and every foreground/background pairing remains legible, including for a visitor using a dark colour scheme or a high-contrast setting.

Why this priority: The operator raised it twice, the second time noting no change had reached the interface. It is P2 because it changes how existing capability is perceived rather than what capability exists.

Independent Test: Load the served interface, sample the colours actually painted on each principal surface, and measure both the variety across them and the contrast within each pairing.

Acceptance Scenarios:

1. **Given** the served interface, **When** its painted colours are sampled across principal surfaces, **Then** they span a materially wider range of distinct hues than a two-tone scheme.
 2. **Given** any text on any surface, **When** its contrast against its background is measured, **Then** it meets the normal-text accessibility floor; non-text indicators meet the non-text floor.
 3. **Given** the interface in light and in dark presentation, **When** both are measured, **Then** both satisfy the two criteria above independently.
 4. **Given** a colour decision recorded in the design source, **When** the served interface is measured, **Then** the served interface reflects it — a measurement taken on the design source alone does not satisfy this story.
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User Story 6 - Each learning module runs on its own (Priority: P3)

An engineer clones either learning module by itself, follows its own written instructions on a machine that has never held this umbrella repository, and reaches a running system.

Why this priority: It affects contributors and deployment rather than learners, and the learner-facing stories can be demonstrated inside the umbrella without it. It is nonetheless a stated requirement and is verifiable.

Independent Test: Clone one module into an empty directory outside the umbrella, run its documented bootstrap, and reach a state its own status command calls ready.

Acceptance Scenarios:

1. **Given** an empty directory outside the umbrella, **When** the engineer clones a module and runs its documented bootstrap, **Then** every dependency the module needs is obtained by that bootstrap.
 2. **Given** the bootstrapped module, **When** the engineer runs its status command, **Then** it reports ready, or reports precisely which host capability is missing and the exact command to obtain it.
 3. **Given** the module in that standalone state, **When** any of its own scripts resolve paths, **Then** none of them require the umbrella to be present.
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Edge Cases

- **An area is authored but incomplete.** It has a title but no lessons, or lessons but no test. It **MUST NOT** appear in the catalogue; a listed area that cannot be completed is defect D3 in a new costume.
- **A test is taken twice.** The result of the submitted attempt is the one that stands; a second submission does not silently overwrite the first without the learner being told.
- **A learner requests a test before finishing the lessons.** The response states what remains. It **MUST NOT** include the questions, the options or the answers — an “unavailable” response that ships the answer key has disclosed it.
- **A referenced recording moment lies beyond the end of the recording.** The reference is reported as unresolvable at authoring time, not silently clamped to the end at reading time.
- **Two areas claim the same subject.** The catalogue exposes each subject once; duplicates are a defect of the authored set and are detectable before publication.
- **The corpus is edited while the catalogue is being measured.** Any count reported about the corpus must be accompanied by evidence that the corpus did not move during the measurement; a count taken over a moving corpus is not reproducible and **MUST** be reported as undetermined rather than as a number.
- **A material file is missing.** The lesson states the material is unavailable rather than rendering a broken element.
- **A visitor uses reduced-motion or high-contrast settings.** Both are honoured; neither degrades legibility below the stated floors.
- **The interface is measured but the served bundle was not rebuilt.** Any interface claim must establish that what was measured is what is served.

Requirements (*mandatory*)

Functional Requirements

Authored subject matter

- **FR-001:** The catalogue **MUST** present knowledge areas that are **authored subjects** — a named topic with a scope a practitioner would recognise — and **MUST NOT** present entries derived solely by extracting a term from source material.
- **FR-002:** Every published area **MUST** be an AI or information-technology subject. The **evidence** for that character **MUST** be

recorded and reviewable per area. The **topical judgement itself** — whether the matched evidence is genuinely about the subject — is a named human checkpoint, not an automated classification.

AMENDED 2026-09-07 — see *Amendment A2*, which is provisional and awaits an operator decision.

- **FR-003:** Every published area **MUST** carry a human-readable title, a summary describing what it covers, and at least one tag. Titles **MUST NOT** be, or contain, an opaque machine identifier.
- **FR-004:** Every published area **MUST** expose the descriptive fields that the established interviewing module exposes **and that the shared curriculum library can express**, so the two present consistently to a learner. Fields the library cannot express are recorded as a named exception list with the reason for each, and **MUST NOT** be fabricated to satisfy this requirement. **AMENDED 2026-09-07** — see *Amendment A3*.
- **FR-005:** The set of published areas **MUST** be derived from the recorded source material's actual AI/IT content, and the coverage relationship between the source material and the published set **MUST** be recorded so gaps are visible.
- **FR-005a:** The published set **MUST** be the authored subject set, and **MUST NOT** be selected by ranking areas on how often their terms occur. Evidence volume **MUST** be reported per area, including for thinly-evidenced areas, rather than used as a publication filter.
- **FR-005b:** An area whose supporting evidence is thin **MUST** still state its evidence volume rather than presenting itself as equally supported.

Lessons and assessment

- **FR-006:** Every published area **MUST** carry at least one lesson, and lessons **MUST** have a defined order.
- **FR-007:** Every lesson **MUST** carry a title, body content and an estimated reading time, and **MUST** be navigable to its neighbours without returning to the area.
- **FR-008:** Every published area for which an authored question bank exists **MUST** carry an end-of-area test drawn from senior-level interview questions on that subject, and an area with no such bank **MUST** carry no assessment rather than a generated one. **AMENDED 2026-09-07** — see *Amendment A1* below.
- **FR-008a:** Assessment questions **MUST** be authored against this corpus. Framing and structure may be adopted from the established interviewing module; question text, options and answers **MUST NOT** be imported from it.

- **FR-008b:** Every assessment MUST declare a pass threshold, and that threshold MUST be shown to the learner before the attempt begins.
- **FR-009:** A learner MUST be able to submit answers to a test and receive a score together with, for each question, whether their answer was correct and what the correct answer is.
- **FR-010:** Until an area's lessons are complete, its test MUST be reported as unavailable with the remaining requirement stated, and the response MUST NOT contain question text, options, or correct answers.
- **FR-011:** A submitted test result MUST be stable — re-reading it returns the same result, asserted by an automated check rather than by a manual validation step.
- **FR-012:** Answers MUST be identified in a way that does not depend on the order options are presented in.
- **FR-012a:** Test availability MUST be enforced where the content is served, not where it is displayed. A learner record held only in the browser MAY decide what is shown; it MUST NOT be what prevents an answer key from being sent.

Materials and video anchors

- **FR-013:** A lesson MUST be able to carry supporting materials — illustration, diagram, scheme, graph, and video segment — each with a caption and a text alternative.
- **FR-014:** A lesson MUST be able to reference a moment or a range in a recorded session, identifying the recording, a start position, optionally an end position, and the transcript passage that corresponds to it.
- **FR-015:** Following such a reference MUST position the recording at the referenced start, scroll the transcript so the referenced passage is in view, and visually distinguish that passage.
- **FR-016:** Every reference in every published lesson MUST resolve to an existing recording and an existing passage; an unresolvable reference MUST prevent publication of that lesson rather than surfacing to a learner.
- **FR-017:** A reference that cannot be satisfied at reading time MUST tell the learner the target is unavailable rather than presenting an empty or not-found page.

Catalogue integrity

- **FR-018:** An area MUST appear in the catalogue if and only if its detail view, its lessons and its assessment are all servable. Listing and openability are one condition, not two.

- **FR-018a:** The publication decision **MUST** be computed in exactly one place and consumed by every surface that needs it. Sharing a vocabulary of reasons between separately-implemented decisions does **NOT** satisfy this — the measured cause of D3 was two implementations that shared their reason strings and differed in their *sequence of tests*, each passing its own unit tests because nothing asserted the pair.
- **FR-018b:** An area withheld from the catalogue **MUST** still be identifiable — its identifier and the reason it was withheld **MUST** remain retrievable, and where a name is derivable it **MUST** be disclosed. Withholding an area is not a reason to withhold what it is called.
- **FR-018c:** An inconsistent build **MUST** be distinguishable from an editorial refusal, and one bad record **MUST NOT** take the catalogue down.
- **FR-019:** Descriptive fields present in the catalogue listing **MUST** survive to the interface that presents them; a field carried by the listing and discarded before display is a defect.
- **FR-020:** The catalogue **MUST** expose each subject once.

Presentation

- **FR-021:** The served interface **MUST** present a palette spanning a materially wider range of distinct hues than a two-tone scheme, across its principal surfaces.
- **FR-022:** Every text pairing in the served interface **MUST** meet the normal-text contrast floor and every non-text indicator the non-text floor, in both light and dark presentation.
- **FR-023:** Design decisions recorded in the design source **MUST** be reflected in the served interface; a claim about the interface **MUST** be supported by a measurement of what is served.
- **FR-023a:** The served catalogue **MUST** declare the corpus revision it was built from, and any claim about served content **MUST** cite that revision. A verified on-disk state is not evidence about what is being served.

Independence

- **FR-024:** Each learning module **MUST** be cloneable and runnable on its own, outside this umbrella, obtaining every dependency through its own documented bootstrap.
- **FR-025:** No module script **MUST** require the umbrella to be present in order to resolve its own paths.

Evidence

- **FR-026:** Every requirement above MUST be covered by an automated check that produces machine-readable evidence.
- **FR-027:** Every such check MUST report three outcomes — satisfied, a real finding, or could-not-determine — and a could-not-determine MUST NEVER be recorded as satisfied.
- **FR-028:** Every such check MUST ship a paired demonstration that it detects the defect it exists to catch, and that demonstration MUST work by supplying different data rather than by altering the check.
- **FR-029:** Every check whose subject is a set MUST establish that the set is non-empty before reporting that the set contains no defects; an absence reported over an empty set is not evidence.
- **FR-030:** Any count reported about the corpus MUST be accompanied by evidence that the corpus did not change during the measurement.

Key Entities

- **Area** — a named AI/IT subject. Has a stable identifier, a human-readable title, a summary, tags, an ordered set of lessons, and exactly one assessment. Published or not; publication requires all its parts to be servable.
- **Lesson** — one unit of teaching within an area. Has a title, ordered position, body content, estimated reading time, zero or more materials and zero or more video anchors.
- **Material** — a supporting artefact attached to a lesson: illustration, diagram, scheme, graph or video segment. Has a kind, a caption and a text alternative.
- **Video anchor** — a reference from a lesson into recorded material: which recording, start position, optional end position, and the corresponding transcript passage.
- **Assessment** — the end-of-area test. Has a question bank, a pass threshold, and an availability condition tied to lesson completion.
- **Question** — one item in an assessment. Has prompt text, identified options, one or more identified correct answers, and an explanation shown after submission.
- **Attempt** — a learner's submission against an assessment. Has the answers given, a score, and per-question outcomes.
- **Catalogue** — the served set of published areas. Its membership is defined by servability, not by a separate flag.

Amendments

Recorded rather than silently applied, because a specification that changes without saying so is as untrustworthy as one that is wrong. Each amendment names the analysis finding that produced it and the measurement that settled it.

Amendment A1 — assessments are required where a bank exists, not universally

Finding F1 (CRITICAL). As originally written, FR-008 and SC-004 required every published area to carry an end-of-area test. The delivered design deliberately permits an area with lessons and no assessment, and the measured state is **5 of 42**. The two positions could not both hold.

Resolved in favour of the design, and the reason is the point. All 44 authored bank questions carry their own area identifier and every one names a platform area; the 37 curriculum areas genuinely have no bank. The alternative — generating 37 banks to satisfy a number — is forbidden elsewhere in this same specification and by the constitution’s *Quality Over Speed* principle, and the validator raises a finding for each missing assessment rather than passing over it. **A fabricated question bank is worse than a missing one: it is a test a learner can fail for the wrong reason.**

The obligation did not weaken. It moved: bank coverage is now **reported on every run** (SC-004a) instead of being asserted and quietly unmet.

Amendment A2 — topical judgement is a checkpoint, not a classifier (PROVISIONAL)

Finding F3 (HIGH). FR-002 required the AI/IT character of each area to be “determinable from a recorded, reviewable classification”. The publication reviews record what was checked **and explicitly what was not** — including *whether the matched evidence is topical*. The requirement therefore asked for exactly the judgement the reviews decline to make.

Provisionally resolved by splitting evidence from judgement: the evidence is recorded and reviewable; the judgement is a named human checkpoint.

This amendment awaits an operator decision and is the only one that does. The alternative — build an automated topical classifier and give it its own review — is defensible and was not chosen here, because it is a curriculum call rather than an engineering one. Two measurements bear on it: a term-in-name probe scores 64.3% AI-or-IT, and its own author records that it *understates* the result because it cannot see areas whose titles carry no jargon. An automated classifier would inherit that blind spot; a human reading the area does not.

Amendment A3 — field parity is bounded by what the library can express

Finding F6 (MEDIUM). FR-004 required exposing “the same descriptive fields” as the interviewing module. Four cannot be expressed here: difficulty and category (no such field exists in the shared library — an upstream change), spaced repetition, and localisation. **As written the requirement was unsatisfiable**, and the only way to “satisfy” it would have been to invent fields.

Two of the four are worth recording precisely, because they are not the same kind of gap. difficulty/category are a genuine capability gap needing an upstream change. Spaced repetition is not: the interviewing module **writes those fields and reads them from no route**, so porting them would add a maintenance surface with no behaviour.

Related and deliberately **not** ported: that module’s kind filter passes an unvalidated value through to storage, so an unknown kind returns an empty list that reads as “there are none”. Here an unknown kind is a **400**. Copying the behaviour would have copied a defect.

Success Criteria (*mandatory*)

Measurable Outcomes

- **SC-001: 100%** of areas visible in the catalogue can be opened; the count of listed-but-unopenable areas is **0**, against a measured baseline of **817**.
- **SC-002: 100%** of published areas are classified AI/IT by a recorded classification, against a measured baseline of **3.9%**.
- **SC-003: 0** published areas display an opaque machine identifier as their title, against a baseline of 3 of 4 render branches doing so.
- **SC-004: 100%** of published areas carry at least one lesson; the count with none is **0**, against a baseline of all of them. **100%** of published

areas that have an authored question bank carry exactly one assessment. **AMENDED 2026-09-07** — see *Amendment A1*.

- **SC-004a:** Question-bank coverage is **reported on every run** as a fraction of published areas, currently **5 of 42**. This is a tracked figure, not a floor: it rises only when a bank is authored against evidence, never by generation.
- **SC-005:** **100%** of published areas carry a summary and at least one tag, and those fields are visible in the interface — not merely present in the data.
- **SC-006:** A learner can go from opening the catalogue to receiving a scored test result **without leaving the product** and without encountering a not-found response at any step.
- **SC-007:** **100%** of video references in published lessons resolve to an existing recording and an existing passage; unresolvable references number **0**.
- **SC-008:** Following a video reference positions the recording within **1 second** of the referenced moment and brings the referenced passage into view without further scrolling, in **100%** of sampled references.
- **SC-009:** Requesting an unavailable test discloses **0** question prompts, **0** options and **0** correct answers.
- **SC-010:** The served interface presents at least **6** distinct hue families across its principal surfaces, up from **2**.
- **SC-011:** **100%** of text pairings meet the normal-text contrast floor and **100%** of non-text indicators meet the non-text floor, in both light and dark presentation; failures number **0**.
- **SC-012:** Each learning module reaches a self-reported ready state from a clone into an empty directory outside the umbrella, on a host that has never held the umbrella.
- **SC-013:** **100%** of the requirements in this specification map to at least one automated check, and every such check ships a paired demonstration that it detects its own defect.
- **SC-014:** **0** checks report a could-not-determine as a pass.
- **SC-015:** A learner completing one area's lessons and test spends no time waiting on a page that has not told them what it is doing.
- **SC-016:** **100%** of published areas report their supporting evidence volume, including the thinly-evidenced ones; areas presenting no evidence figure number **0**.
- **SC-017:** **100%** of assessments declare a pass threshold that is visible before the attempt begins.
- **SC-018:** The served catalogue declares the corpus revision it was built from in **100%** of responses, so a served-versus-on-disk discrepancy is detectable rather than silent.

Assumptions

These are reasonable defaults chosen where the input did not specify. Each is recorded so it can be challenged in clarification rather than discovered later.

- **Learners are anonymous** (decided, Q2). Progress and attempts are held per browser session; the serving side, not the browser, is what withholds an unavailable test's content. Durable cross-device progress is a separate capability.
- **Content is authored in English.** The recorded source material and the existing lessons are English; localisation is out of scope for this feature.
- **The existing interviewing module is the reference for field shape,** because the operator named it as the model. It is a reference for *what fields exist and how they present*, not a source of content to copy.
- **“Senior-level” means questions a practising engineer with several years in the subject would find non-trivial** — scenario and trade-off questions rather than definition recall. The bank is authored against this corpus (decided, Q3), not generated from transcript text and not imported from the interviewing module.
- **Video is served from material already present in the private module.** No new recording, hosting or transcoding capability is introduced.
- **Materials are authored assets committed alongside the lessons.** No authoring tool or upload path is introduced by this feature.
- **The palette is derived from the existing seeded design tokens** rather than chosen afresh, so brand continuity holds; what changes is that the derivation reaches the served interface.
- **Existing search, answer and evidence capability is unchanged.** This feature adds a curriculum surface beside it and must not regress it.
- **Scale is 37 authored subjects over three recorded sessions** (decided, Q1), matching what the corpus holds today. The design must not assume those numbers, but the acceptance evidence is measured against them.
- **Publication is a build-time determination.** Whether an area is servable is decided when the catalogue is assembled, not per request.

Dependencies

- The private curriculum corpus and its recordings, which are referenced by path and never by content.
- The existing transcript passage registry, which supplies the passages video anchors resolve against.
- The existing design token source, from which the palette is derived.
- The reusable curriculum library that carries the area/lesson/material/assessment shape, so this capability is not welded to one application.

Out of Scope

- Learner accounts, cross-device progress, and certificates.
- Authoring tools or a content management interface; content is authored as committed files.
- Localisation of lessons or questions.
- New recording, video hosting or transcoding capability.
- Changes to the existing search and answer surfaces beyond what integration requires.